

Cameron Ake

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EDUCATION

Harvey Mudd College

B.S. Computer Science and Mathematics – Cumulative GPA: 4.00 / 4, Dean's List

Claremont, CA

Expected May 2028

Thomas Jefferson High School for Science and Technology

High School Diploma – GPA: 4.629 / 4, National Merit Scholar

Alexandria, VA

June 2024

INTERNSHIP EXPERIENCE

Data and Software Engineering Intern, Fermat Capital Management

May 2026 – Aug. 2026

- Automating quantitative risk analysis pipelines to surface exceedance probability curves from databases on demand, replacing manual querying workflows for quantitative reinsurance team
- Developing full-stack web application to automate R&D workflows, integrating Anthropic models via AWS Bedrock and Gmail, Drive, and Box APIs

Researcher, Harvey Mudd College Dept. of Mathematics

May 2025 – Aug. 2025

- Modeled and analyzed quantitative stack-sorting polytope attributes using SageMath in Python
- Proved all stack-sorting polytopes are simplices and all $Ln1$ polytopes have relative volume 1
- Co-authored final paper published in the American Journal of Combinatorics (doi.org/10.63151/amjc.v4i.35)

AI Engineering Intern, M.C. Dean

June 2023 – Aug. 2023

- Built LSTM models to predict category and priority values for database inputs
- Presented and compiled final report to Chief Systems Engineer, recommending integration into database UI

PROJECT EXPERIENCE

Ticket to Ride Bot | cameronake.com/ttr-demo

- Engineered custom graph structure, helper algorithms, and heuristics for bots to play the game *Ticket to Ride*
- Developed full-stack web application to set hyperparameters for GA and breed bots from scratch
- Created bot that typically performs 20 points better than the game publisher's best bot

Bayesian Optimization | cameronake.com/bo-demo

- Created full-stack Bayesian optimization web demo built from scratch (only using NumPy math operations)
- Used GPR with user-set target black-box function, parameters, and acquisition function

Monte Carlo Options Pricing | cameronake.com/mcop-demo

- Created web demo of Black-Scholes and Monte Carlo methods to price options based on user-set parameters
- Simulated GBM on stock prices to compute average payoff of option and compare to Black-Scholes computed price

LEADERSHIP EXPERIENCE

Mentor & Knott's Head, Harvey Mudd College Residential Life

Apr. 2025 – Present

- Delivering structured support for freshmen through transition to college, as part of a team of 5
- Planning programming, including leading organization of freshman class Knott's amusement park trip

Grader & Tutor, Harvey Mudd College Dept. of Computer Science

Aug. 2025 – Present

- Grading assignments and tutoring: Intro CS (Python); Data Structures and Program Development (C++)

SKILLS

Languages: Python, Java, C/C++, SQL, JavaScript, HTML/CSS, R, MATLAB, Racket, Prolog

Libraries: TensorFlow, PyTorch, NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, SQLAlchemy

Tools: Git, Regex, REST APIs, LaTeX, Excel, GDB, Jamovi, Node.js, Pyodide, Claude Code, VS Code, Jupyter

Coursework: Bayesian Statistics, Real Analysis, Complex Analysis, Machine Learning, Artificial Intelligence, Data Structures/Program Development, Computability and Logic, CS Principles and Practice, Computer Systems Research, Probability and Statistics, Differential Equations, Discrete Mathematics, Linear Algebra, Multivariable Calculus